**SmartBridge** # Solution Architecture **SKILL WALLET**

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|---|---|
| **Date** | 27 June 2026 |
| **Team ID** | xxxx |
| **Project Name** | A Comprehensive Measure of Well-Being |
| **Maximum Marks** | 3 Marks |

## Solution Architecture Diagram

A high-level view of the HDI prediction app's architecture, mapped onto the standard Presentation, API, Service and Data layers.

**Presentation / Client Layer**
HDI Prediction Web App (Flask + HTML forms) — home.html, indexnew.html

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**API Gateway / Load Balancer**
Flask application routes acting as the API layer (Nginx reverse proxy in production)

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| **Auth Service** | **Core Logic Service** | **External API Integrations** |
|---|---|---|
| Not required for general use; optional admin login for model management | HDI Prediction Service — loads model.pkl and classifies indicator input (scikit-learn) | Optional: World Bank / UNDP open data API for live indicator lookups |

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|---|---|---|
| (Flask-Login) | | |

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| **Data / Storage Layer**<br>Model store for model.pkl; optional database for logging past predictions |
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**Component Description Table**

| | | |
|---|---|---|
| Presentation Layer | Web page where users enter indicator values and view the predicted HDI category. | Flask, HTML/Jinja templates (home.html, indexnew.html, resultnew.html) |
| API Gateway | Routes incoming requests (e.g. /predict) to the right backend handler. | Flask routing |
| Auth Service (optional) | Protects model-management actions for an admin user; not required for general predictions. | Flask-Login |
| Core Logic Service (ML) | Loads the trained model and predicts the HDI category from the submitted indicator values. | scikit-learn, joblib, pandas, numpy |
| External API Integrations (optional) | Could pull live indicator data from open data sources instead of manual entry. | World Bank / UNDP open data APIs |
| Data / Storage Layer | Holds the serialized model and can log past predictions for later review. | model.pkl (file storage); optional SQLite/PostgreSQL log |